

First steps towards novel adaptive systems for individualised music learning and assessment: An investigation of memory chunking in a piano chord production task

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Abstract. This paper presents a simple pilot study that uses cognitive behavioural measures to assess piano playing proficiency. The wider objective is to explore how digital game-based learning environments could be designed to adapt to learner behaviour in the musical domain. We believe it is possible to utilise cognitive behavioural metrics to not just assess aspects of musical proficiency but more importantly, to drive the selection and configuration of suitable learning sessions as well as teaching methods and content. With our approach, we aim to create a learning paradigm that adapts dynamically to a learner and avoids overly rigid instruction approaches that focus on the precise replication of musical patterns, which can make it difficult for learners of music to experience progress and stay motivated. Instead, we propose that cognitive behavioural metrics could be used to inform learning trajectories that give insight into learning progress through changes observed in the cognitive processes that underlie learning. In this paper, we present the results of a pilot experiment that compares the memory chunking signals obtained from participants of different musical proficiency who engaged in a musical chord reproduction task. The purpose of this experiment was to establish if results from previous research that targeted other domains of learning (e.g., maths, writing, drawing, etc.) are, in principle, transferable to the music domain. The results found indicate that this is likely the case.

1 INTRODUCTION

Pedagogies that assess music learners solely based on task accuracy promote a rote approach to learning - being only concerned with the precision of outputs rather than the indication of a deeper understanding of the target domain. For example, accuracy may be a suitable metric for classical pianists who seek to strictly maximise execution fidelity of a particular musical piece. In contrast, many jazz pianists will routinely use improvisation to enrich their performance, and these musicians must associate knowledge between various other aural and conceptual aspects to be proficient in their area. Accuracy cannot capture all elements of improvisation, nor of piano ability in general, as the ability to interpret music is also an indicator of proficiency and hence, there rarely is a single standard for comparison. A more cognitively informed digital learning environment could relate aspects of task performance to relevant cognitive functions, capable of constructing a higher-dimensional model of learner ability to

identify areas of weakness that may be limiting performance in other areas.

2 RELATED WORK

Interpreting behavioural measures to study mental processes is a long-established method of investigation in Cognitive Science, and a large body of work exists that has investigated memory chunking in the context of different tasks. Specifically, the pauses that are made by individuals when they are performing a task were found to coincide with information retrieval and processing activities, and the temporal patterns of pauses between actions can give insight into how information is organised in human memory. We also know that the size of memory chunks is related to our ability to retain different amounts of information, and that chunking structure also reflects the processes underlying working memory management and can indicate the strength of association between memory contents. [2][4].

Experts and novices structure information in working memory differently. Given the same task, experts are generally able to store more information per chunk than novices and related tasks induce less cognitive load [10][16]. The amount of cognitive load experienced can be estimated by measuring behavioural patterns, such as the timings of a student's actions. Student proficiency for a given task could therefore be measured by observing how their behavioural patterns compare with those typically observed in experts and novices [9][14]. Although there can be individual differences between people with equivalent degrees of proficiency, previous research has shown that similarities within their behavioural patterns can be extracted and analysed. For example, Cheng and Rojas-Anaya [6] were able to successfully categorise the mathematical competency of participants by analysing data about how they wrote down maths formulae: their writing actions, pen positions and timings, and the duration of pauses between drawn elements. Since playing piano involves a similar externalisation of intended actions through hand and finger movements, there is a distinct possibility here that chunking could contribute towards establishing a more comprehensive measurement of musical proficiency. We contend that one particularly promising way of obtaining such measures may be game-based learning environments, as these permit the collection of data not just over time, but also across different learning activities and contexts. Interpreting an individual learner's chunking pattern could help indicate when internal processing is happening, and point to areas of weakness in their musical skill that other methods might miss.

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Previous work has found strong correlations between proficiency and temporal behavioural metrics that were used to examine individual chunking structures in domains such as mathematics [5], natural language writing and comprehension [11], or computer programming [1]. Recent PhD theses have also examined chunking in the context of people reading musical notation [7] [12]. The pilot study presented in this paper aims to establish if people with different proficiency levels in piano playing might, in principle, exhibit chunking patterns that are consistent with the results previously found in those other domains.

3 APPROACH

The experiment discussed in this paper was conducted on Sussex AI Day in January 2026 at the University of Sussex. The event showcased current ongoing AI projects by research groups across the University. Using the Unity Game Engine [17], custom software was written in advance in C# to carry out the experimental trials and record relevant data during the event. There was a one hour window of time when attendees were able to volunteer to participate. This allowed for opportunistic sampling of participants who belonged to a range of different piano proficiency bands. Before starting the first trial, participants completed a pre-trial questionnaire related to their familiarity, confidence, and experience with the piano, which was used to categorise them each based on their general piano proficiency. The questionnaire had 12 questions, but to keep the categorisation process simple, only one question was used to determine an individual's proficiency group: "To what extent do you play piano?". This is because it would have been difficult to set weights for how each variable contributes to overall proficiency without prior testing. We gathered a total of 14 participants. The distribution of participants across categories is as follows: Complete Beginner (4), Hobbyist (6), Intermediate (3), Professional (1).

The experimental procedure involved 2 trials, preceded by a familiarisation trial, in which participants produced chords from memory immediately after viewing a stimulus of 15 chords, as shown in figure 1. Participants interfaced with the experimental trial using a standard 49-Key MIDI piano keyboard. The stimulus was hidden during chord production but could be viewed again upon request as often as necessary by holding down the space key on the laptop keyboard. Participants were instructed to use the same hand to both view the stimulus and to play chords with. This was to minimise task execution rehearsal while the stimulus was being viewed, since participants could not simultaneously get their hand into the correct position to play the next chord whilst viewing the stimulus.

The specific variables measured were determined by which behavioural metrics we hypothesised would correlate with piano proficiency. Average metric values were calculated for each participant, and then averaged again across proficiencies. They are as follows:

- M1)** Number of chords played between stimulus views
- M2)** Number of stimulus views per trial
- M3)** Stimulus view durations (seconds)
- M4)** Chord-playing durations (play periods) between stimulus views (seconds)
- M5)** Pause durations between chords (seconds)

This selection of metrics allows for the possibility of analysis to occur at a multiple-chunk scale (e.g. number of stimulus views), as well as at the level of the elements within individual chunks themselves (e.g. pauses between chords).

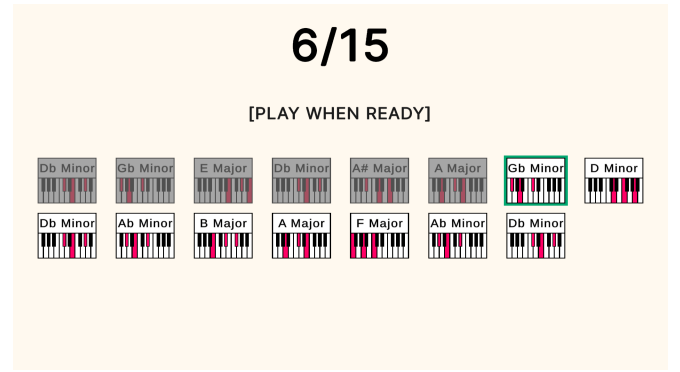


Figure 1. A screenshot from the experimental software, showing the chord progression stimulus for memorization (Trial 2). The next chord to be played is highlighted in green and the chords that have already been played are greyed out. "6/15" at the top refers to the number of chords the participant has played so far, out of 15. When the space-bar is released, the stimulus disappears from the screen.

4 RESULTS

As shown in figure 2, the results showed that higher proficiency was correlated with fewer stimulus views per trial, more chords played between stimulus views, and shorter pauses between chords. Correlations between stimulus view durations or play time durations with proficiency was unclear, which is discussed further in section 5.1. These results are consistent with previous work in other domains which find significant correlations between similar behavioural metrics and proficiency [1]. All descriptions of results are speculative, given that there was insufficient data to form any significant conclusions.

The data of 3 participants was excluded from these results for the following reasons:

1. Participant did not attempt to recreate the stimulus chords in any way and just pressed notes in ascending order.
2. Participant was too engaged in the exact scientific discussion before attempting the trial, which primed them to attempt to memorise far more chords (8-10) at a time than we believe they otherwise normally would have - skewing the results.
3. Participant did not seem to understand that they could re-view the stimulus upon request and attempted to memorise all chords in one try.

5 DISCUSSION & FUTURE WORK

5.1 Interpretation of Results

While further work will need to be done to properly control for these assumptions, this section discusses how the results for each metric (as shown in figure 2) may be explained by chunking processes.

- **M1 & M2):**

The average number of chords played between stimulus views measures how many chordal elements participants were able to recall. Since experts are able to incorporate more elements per chunk [10], we expect to see this metric increase as proficiency increases. For the same reason, we would expect to see the average number of times a participant chooses to view the stimulus

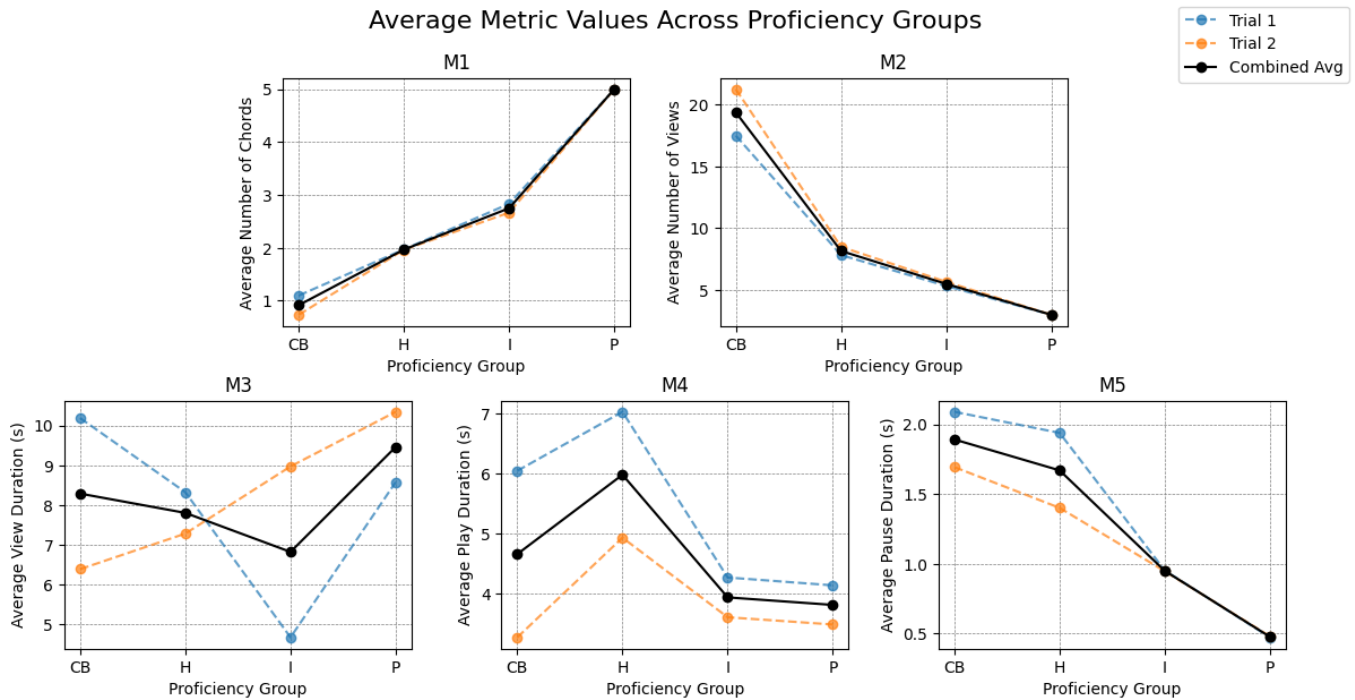


Figure 2. The observed results of all 5 metrics (M1-M5) gathered from behavioural data during both chord production memory trials. Proficiency groups are abbreviated as follows: Complete Beginner = CB; Hobbyist = H; Intermediate = I; Professional = P. Blue dashed lines represent trial 1 results; Orange dashed lines represent trial 2 results; Black solid lines represent combined trial results.

during trial decrease as proficiency increases, since a less efficient working memory will have to be refreshed more often. Both of these trends were observed in the results.

An experiment with similar methods tracked eye movements during a stimulus matching task to determine the conditions under which individuals favour online access - using the environment as its own model to complete the task as opposed to storing multiple elements of the stimulus in memory [3]. They found that participants would refer to the stimulus more if the cost of online access was less than the cost of memorisation. Looking at our study from this perspective, experts may produce more chords from memory because the size of their chunks result in the cost of memorisation being lower than the cost of requesting to view the stimulus. Understanding when individuals use the world as its own model and when they choose to rely on internal models from a chunking perspective may benefit our understanding of domain proficiency in a way that incorporates elements of 4E cognition.

- **M3):**

The average time taken to view the stimulus represents the amount of time it takes for a participant to fill their working memory with relevant chordal information to complete the memory task. The lack of a strong identifiable trend between view durations across proficiencies in the results may be explained by the fact that the time to perceive chunks is approximately constant - regardless of the number of elements within each chunk [4][1]. While analysing the data, we discovered that dividing the stimulus view durations by the number of chords that were memorised from that view provided a way of normalising this metric. Figure 3a shows that experts viewed the stimulus for less time per chord produced, indicating the presence of more efficient perceptual chunking of

chordal elements over novices.

- **M4):**

The average amount of time it takes for participants to produce the chords retained in working memory is likely independently affected by various factors. The participant's domain-specific motor ability will determine how fast they are able to play each chord, and therefore the amount of time to execute each full play period before having to review the stimulus. However, play duration will also be a function of how many chords were memorised as, *ceteris paribus*, it takes more time to play more chords. These confounding factors could explain the lack of a linear trend between play durations and proficiencies. Similarly to **M3**, normalising play period durations according to the number of chords played during that period reveals a decrease in the amount of time taken to play each chord as proficiency increases, as shown in figure 3b. Given that the processing of elements across chunk boundaries is shown to take more time than for elements within the same chunk [2][15], the faster production of chords from experts suggests the presence of larger chunks.

- **M5):**

The pause durations between each chord decrease as proficiency increases - which is a finding that is consistent with chunking theory. Experts likely hold whole chords as single homogeneous chunks that can be executed without further decomposition. Novices have to decompose each whole chord into sub-chunks to execute, which increases processing time, and therefore, pause duration.

While there is not enough data from this pilot study to provide any statistical significance, these results indicate that there may be

valid, operationalisable methods to identify piano proficiency based on analysing chunking structure. Furthermore, the promising trans-ferral to the musical domain of piano playing strengthens the theory behind previously identified works which utilise behavioural metrics to assess proficiency in other domains [5], [11], [1].

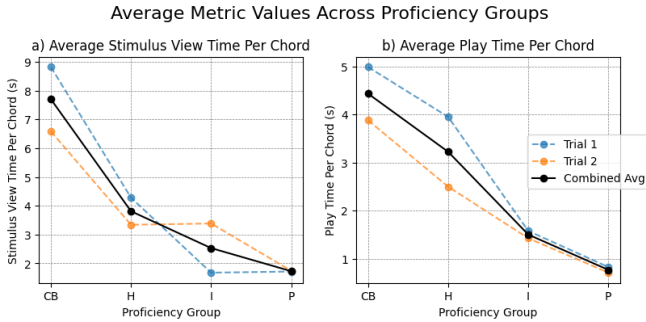


Figure 3. Metrics M3 and M4 normalised per chord: a) Average stimulus view time per chord - created by dividing M3 by M1. b) Average play time per chord - created by dividing M4 by M1. Proficiency groups are abbreviated as follows: Complete Beginner = CB; Hobbyist = H; Intermediate = I; Professional = P. Blue dashed lines represent trial 1 results; Orange dashed lines represent trial 2 results; Black solid lines represent combined trial results.

5.2 Limitations

5.2.1 Participant Data

Due to the small window of time that was assigned for the opportunistic sampling of participants, there were several issues regarding how data had to be collected, which prevented meaningful statistical analysis. The sample size was not large enough, nor was the distribution of participants across proficiency levels able to be controlled for. Factors that may have confounded with the observed results could also not be controlled for. This includes components that affect working memory access, such as age and various neurodivergent profiles, which are likely to influence individual task performance. Pre-trial proficiency evaluations could not be objectively determined, and relied on subjective answers from the given questionnaire. All observed trends are therefore merely indications of promising areas for future research. It is also worth noting that the public nature of the event may have introduced attentional challenges which distracted participants during trials. Future studies extending this methodology should attempt to control for these factors within a greater sample size. Work which evaluates whether the presented method can discriminate within a narrower range of proficiencies (e.g. between only intermediate players) would be particularly valuable for digital learning environment contexts.

5.2.2 Assessment Task Design

Assumptions were made about how the proposed metrics relate to internal cognitive mechanisms and chunking structures. For example, metrics **M1 & M2** assume that each new request to view the stimulus is representative of the fact that all of the chordal elements in working memory have been produced and that new elements need to be loaded in. However, there may be other factors affecting the rate of stimulus

viewing and chord production, such as self-confidence, which does not always align with proficiency (e.g. see the Dunning-Kruger effect [8]). Work needs to be done to thoroughly investigate the cognitive implications of each of the proposed metrics to increase their validity. This also includes more strict control over the difference in potential strategies employed by participants to ensure that behavioural metrics are cohesively measuring the same chunking phenomena.

The presented assessment method is ultimately only a proposed measure of the chord production element of piano playing, which is a subset of the skills required for an individual to be considered a skilled piano player. Further work will need to investigate how this kind of task can be transferred to other aspects of piano playing, such as sight-reading or scale production. As some of the proposed metrics rely on measuring temporal signals of behaviour, challenges may arise when applied to rhythmic aspects of piano playing which determine the exact timing of inputs required. All individuals who are able to match the target rhythm exactly will produce the same temporal signal, producing a ceiling effect on the data which would obfuscate the distinction between their levels of proficiency above a certain point. One way to circumvent this issue may be to gradually increase the difficulty of the stimulus, or the required tempo of the response - producing more variation in results between individuals by raising the ceiling effect. However, it is important to keep note that the more obscure the assessment method, the less likely the target of that assessment may align with the target skill - this is also a necessary direction for future research and is discussed further at the end of section 5.3.

5.2.3 Stimulus Design

Currently, no consideration has been made to investigate the effect that specific components of the stimulus have on resulting chunking signals. For example, experts may perform better when the stimulus content is presented according to common harmonic patterns, whereas this may not affect novice performance. There will also be variation as to the specific chords/ chord types that are easier to produce between individuals. Observing how idiosyncratic task performance changes in response to various stimulus components will indicate the strength of association that individual learners have with task-relevant musical elements. Research can then be directed towards effective ways to tailor the sequencing of learning content given these kinds of qualitative models of player proficiency. Human-Computer Interaction methods may also be able to reveal design implications for producing more reliable chunking signals by reducing the extraneous cognitive load [13] caused by visual elements of the stimulus and its layout.

5.3 Motor Skill vs. Conceptual Knowledge

The pilot study does not specifically distinguish between conceptual domain knowledge and motor ability. For example, metrics **M4 & M5** suggested that the average time taken to play each chord may be predictive of proficiency. However, these metrics are likely determined by both the conceptual knowledge of what notes are in each chord and the motor ability to efficiently configure the necessary body parts to physically play the chord - both could be viewed as distinct proficiencies. Given this, it is initially unclear how much of the chunking signal is distorted by complex interactions between motor execution and conceptual knowledge. Future work aimed at disentangling this interaction may be beneficial to clarify the validity of the proposed metrics, and for informing decisions related to

how digital learning environments use these metrics to adaptively sequence individualised learning content.

It is also worth considering the degree to which this distinction is necessary; studies from embodied cognition highlight the key involvement that sensorimotor loops have in working, implicit, and long-term memory (see [18] for a review). Since motor processes are seemingly involved throughout many key stages of cognitive processing, a more holistic approach to skill acquisition could involve purposefully blurring the lines between conceptual knowledge and motor skill. This is specifically pertinent to the domain of piano playing. For example, previous work has attempted to minimise for the variation of motor skill because the method of input is unrelated to the domain knowledge being measured (e.g. differences in typing ability would interfere with the signal when testing for programming knowledge [1]). For piano playing, however, more properties of the chunking signal may be able to be utilised, as it is directly related to the target skill - i.e. playing the piano. The transferral of chunking-based proficiency evaluation to this domain may be therefore significant because it suggests that the proposed method is robust enough to accommodate for, and even take advantage of, large variations in task-dependent motor skill between experts and novices.

In a similar vein, future research can investigate how the quality of proficiency measurement can depend on the alignment between an assessment method and the target of assessment. For example, to gain chunking metrics, the proposed assessment requires participants to complete a memory task that they would likely not encounter during normal piano playing scenarios: producing a long, unseen chord sequence from memory without a defined tempo. It is important that assessment tasks are designed in such a way that they produce measurements of the appropriate skills pertaining to the target learning objective, and not some kind of arbitrary skill that is too abstract and thus irrelevant to general piano playing.

6 CONCLUSION

We envision digital learning environments that operate in what appears almost like a niche in today's AI landscape, where hyper-automation is often the most obvious - but not necessarily most useful - approach: Systems that support humans in performing their tasks in a collaborative fashion rather than systems that are trying to replace them entirely. By using chunking as a framework for informing the mechanics of digital learning in creative domains, the proposed method of digital assessment could employ behavioural metrics to model piano players' skill based on the richness of associations that they have with musical material, as opposed to the accuracy with which they can play specific material. We hope that this will be useful to augment piano tuition and aid creativity and skill in a way that does not require algorithms to make overly rigid value judgements in what is largely an aesthetic domain, such as determining what may or may not be a "good" jazz improvisation, for example. Beyond this, we believe that chunking metrics could be useful for informing AI models that could drive the selection and presentation of learning content in assistive technologies, to more flexibly adapt to student proficiency, and with the aim to yield more effective tuition and support long-term motivation. While these systems are currently a long way off, the results of this study have shown promising correlations between the proposed metrics and observed proficiencies, and clearly motivate further research in this area.

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REFERENCES

- [1] Noorah Albehaijan and Peter C.-H. Cheng, 'Measuring Programming Competence by Assessing Chunk Structures in a Code Transcription Task', *Proceedings of the Annual Meeting of the Cognitive Science Society*, **41**(0), (2019).
- [2] John R. Anderson, *Learning and memory: an integrated approach*, Wiley, New York, N.Y., 2nd ed. edn., 2000.
- [3] Dana H. Ballard, Mary M. Hayhoe, Polly K. Pook, and Rajesh P. N. Rao, 'Deictic codes for the embodiment of cognition', *Behavioral and Brain Sciences*, **20**(4), 723–742, (December 1997).
- [4] William G. Chase and Herbert A. Simon, 'THE MIND'S EYE IN CHESS', in *Visual Information Processing*, ed., WILLIAM G. CHASE, 215–281, Academic Press, (1973).
- [5] Peter C.-H. Cheng, 'Analyzing chunk pauses to measure mathematical competence: Copying equations using 'centre-click' interaction', *Proceedings of the Annual Meeting of the Cognitive Science Society*, **37**(0), (2015).
- [6] Peter CH Cheng and Hector Rojas-Anaya, 'Measuring mathematic formula writing competence: An application of graphical protocol analysis', in *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 29, (2007).
- [7] Rachel Wooryung Dueck, *Developing a Progressive Chunking Technique in Music Literacy: An Analysis of the Method Books and Sight-Reading Materials for Beginner Piano Students*, Ph.D. dissertation, Université d'Ottawa / University of Ottawa, November 2023.
- [8] David Dunning, 'Chapter five - The Dunning–Kruger Effect: On Being Ignorant of One's Own Ignorance', volume 44 of *Advances in Experimental Social Psychology*, 247–296, Academic Press, (2011).
- [9] Dennis E Egan and Barry J Schwartz, 'Chunking in recall of symbolic drawings', *Memory & cognition*, **7**(2), 149–158, (1979).
- [10] Fernand Gobet, Peter C R Lane, Steve Croker, Peter C-H Cheng, Gary Jones, Iain Oliver, and Julian M Pine, 'Chunking mechanisms in human learning', Technical report, (2001). Publication Title: TRENDS in Cognitive Sciences Volume: 5 Issue: 6.
- [11] Hadeel Bakr M. Ismail, *Competence assessment by stimulus matching (CASM): a novel approach to language assessment by chunk transcription*, thesis, University of Sussex, March 2025.
- [12] Lucas Lörch, Expertise in processing musical notation. An investigation of chunking, working memory, and eye movements, 2023. Place: Mannheim.
- [13] Richard E. Mayer, *Multimedia Learning*, Cambridge University Press, July 2020.
- [14] Robert S McLean and Lee W Gregg, 'Effects of induced chunking on temporal aspects of serial recitation.', *Journal of experimental psychology*, **74**(4p1), 455, (1967).
- [15] Timothy P. McNamara, James K. Hardy, and Stephen C. Hirtle, 'Subjective hierarchies in spatial memory', *Journal of Experimental Psychology: Learning, Memory, and Cognition*, **15**(2), 211–227, (1989).
- [16] George A. Miller, 'The magical number seven, plus or minus two: Some limits on our capacity for processing information', *Psychological Review*, **63**(2), 81–97, (1956). Place: US.
- [17] Unity Technologies. Unity game engine, 2026. <https://unity.com>, Accessed: 2026-08-06.
- [18] Margaret Wilson, 'Six views of embodied cognition', *Psychonomic Bulletin & Review*, **9**(4), 625–636, (December 2002).